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% Study of digital to analog conversion
close all;
clc;

f1 = 3;
f2 = 10;
fc = 5;

x = [1 1 0 0 1 0 1 0];
nx = size(x,2);

i = 1;
while i < nx+1

    t = i:0.001:i+1;
    ct = sin(2*pi*fc*t);

    if x(i) == 1
        ask = sin(2*pi*fc*t);
        fsk = sin(2*pi*f2*t);
        psk = sin(2*pi*fc*t);
    else
        ask = 0;
        fsk = sin(2*pi*f1*t);
        psk = sin(2*pi*fc*t + pi);
    end

    subplot(4,1,4);
    plot(t, ask);
    hold on;
    grid on;
    axis([1 10 -1 1]);
    title('Suraj k. jha Amplitude Shift Key');

    subplot(4,1,2);
    plot(t, fsk);
    hold on;
    grid on;
    axis([1 10 -1 1]);
    title('Suraj k. jha Frequency Shift Key');

    subplot(4,1,3);
    plot(t, psk);
    hold on;
    grid on;
    axis([1 10 -1 1]);
    title('Suraj k. jha Phase Shift Key');

    subplot(4,1,1);
    plot(t, ct);

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hold on;
grid on;
axis([1 10 -1 1]);
title('Suraj k. jha Carrier Signal');

i = i + 1;

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end
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